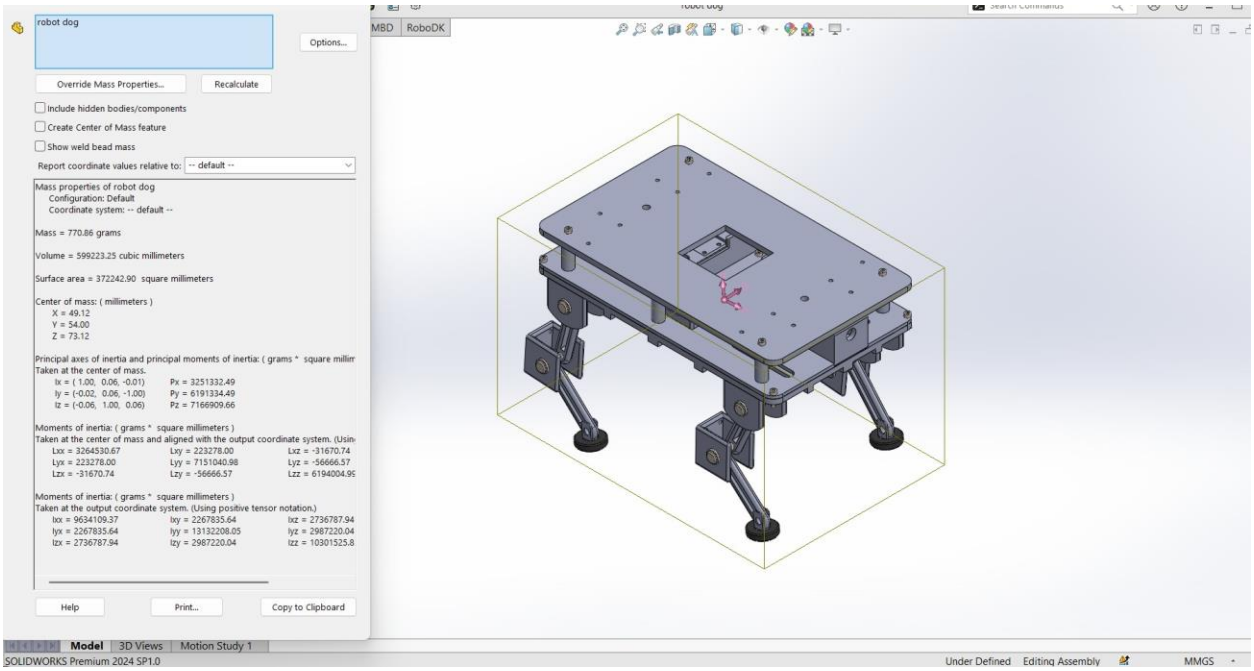
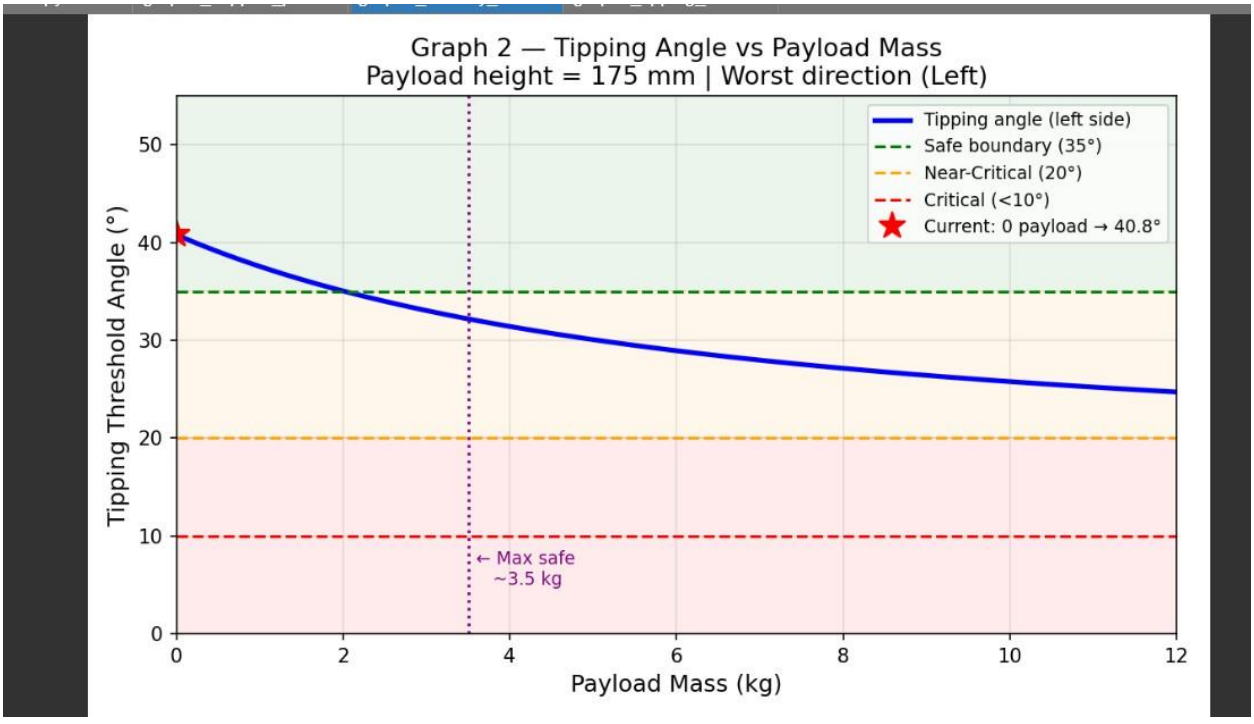
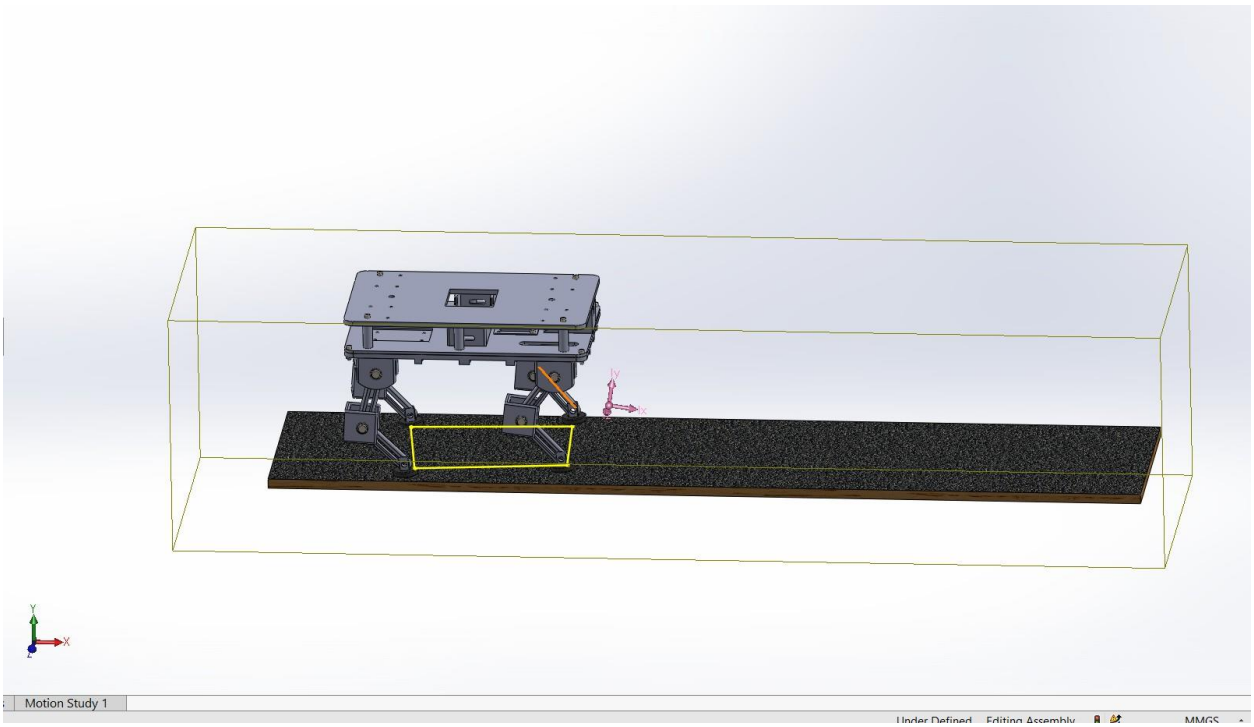
Term	
Resources	SolidWorks, 3 D printer, SolidWorks simulation, JSON code
Progress	CAD model, motion and terrain analysis, gait reasoning, risk register, BOM, BOQ, polygon analysis, production layer, stability, force vs time graphs for each leg, actuator load validation
Blockers	3 D printing of all assembly and its working after assembly
Dependency / Chain	3 D Printing
Approval Dependency	Yes
Partial Ownership	No
Complete Ownership	Yes.
Owner	Arya Barge
Contributor	NO
Next Action	3D Print all the components and assemble them to check the motion analysis and any errors occurs
Evidence	CAD file with all stl packages of each component, force vs time graphs, polygon anlysis screenshot, BOM, BOQ
Status	In progress


robot dog
Options...
Override Mass Properties...
Recalculate
Include hidden bodies/components
Create Center of Mass feature
Show weld bead mass
Report coordinate values relative to: -- default --
Mass properties of robot dog
Configuration: Default
Coordinate system: -- default --
Mass = 770.86 grams
Volume = 599223.25 cubic millimeters
Surface area = 372242.90 square millimeters
Center of mass: (millimeters)
X = 49.12
Y = 54.00
Z = 73.12
Help
Print...
Copy to Clipboard
MBD
RoboDK
Model
3D Views
Motion Study 1
SOLIDWORKS Premium 2024 SP1.0
Under Defined
Editing Assembly
MMGS


Graph 2 — Tipping Angle vs Payload Mass
Payload height = 175 mm | Worst direction (Left)
Tipping angle (left side)
Safe boundary (35°)
Near-Critical (20°)
Critical (<10°)
Current: 0 payload → 40.8°
Tipping Threshold Angle (°)
Payload Mass (kg)
← Max safe
~3.5 kg
0
10
20
30
40
50
0
2
4
6
8
10
12


Y
X
Z
Motion Study 1
Under Defined
Editing Assembly
MMGS